SUPPORTING ADULT GUIDANCE

Circumference of a Circle

2D and 3D Shapes. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner discovers that the circumference of a circle is a fixed multiple of its diameter — the number π (about 3.14) — and uses C = π d to find a circumference. The emphasis is on **seeing π as a constant ratio** (the same for every circle), reached by discovery rather than by being told the formula.

Driving Question: *How far is it around a circle — and how does that distance depend on the circle's width?*

The mathematics, in plain terms

The **circumference** is the distance all the way around a circle; the **diameter** (d) is the distance straight across through the centre; the **radius** (r) is from the centre to the edge, so d = 2r. For *every* circle, the circumference divided by the diameter is the same number, about 3.14, which we call π (pi). So the circumference is **C = π d** (equivalently C = 2 π r). Because C = π d, doubling the diameter doubles the circumference — the two are in direct proportion. If you plot circumference against diameter, the points make a straight line through the origin whose gradient is π. To go backwards, from circumference to diameter, divide by π: d = C ÷ π.

π as a ratio, not just a button

The most useful idea to protect is that π is a *ratio* — the circumference is always about 3.14 diameters — not merely a number on a calculator. A learner who has watched the wheel roll out three-and-a-bit diameters, or seen the plotted points line up, understands *why* C = π d, and can estimate a circumference sensibly ('a bit more than three times across'). The second habit is care with radius and diameter: if a question gives the radius, the diameter is double it, and only then multiply by π. Let the learner reach the equation through the investigation before it is written down for them.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The (d, C) Plotter: plotting points, seeing them line up, and discovering that the gradient — the ratio C ÷ d — is π.	~15 min
Movement	The Rolling Wheel: one full turn unrolls the circumference to about three-and-a-bit diameters — that multiplier is π.	~8 min

Reflection	Judging true/false statements about circumference and π, and giving the reason why.	~10 min
Creativity	Estimating the distance around real objects (a coin, a wheel, the Moon) using C is about 3.14 times d.	~5 min
Rest	A pause to notice which made π feel real — the points lining up, or the wheel rolling out.	~4 min
Nutrition	Contextual (Mode A): the label on a tin of food must be as long as the tin's circumference to wrap right around it.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking the circumference is exactly 3 times the diameter

Repair: “Roll the wheel one full turn — does it stop exactly on the 3-diameter mark, or a little past it?”

Muddling radius and diameter

Using the radius as if it were the diameter. Repair: “Is that measurement across the whole circle, or only halfway? If it is the radius, what is the diameter?”

Believing π changes with the size of the circle

Repair: “Divide the circumference by the diameter for a small circle and a big one — do you get the same number?”

Treating π as a distance

Rather than a ratio. Repair: “π has no units — it just tells us how many diameters fit around. About how many is that?”

Adding instead of multiplying

Repair: “The circumference is π lots of the diameter — is that an add or a multiply?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: multiplying by a decimal, ratio (‘times bigger’), and the names of the parts of a circle. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the reasoning — and the ratio — with the learner:

- “Point to the diameter. Now the radius. How are they related?”
- “About how many diameters fit around the circle — closer to 3, or to 4?”
- “The diameter is 10 cm. Roughly how far is it around, before you calculate?”
- “If this gives the radius, what is the diameter — and then the circumference?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do they need to remember 3.14, or use the π button?”
Either is fine, as long as they understand π is the fixed ratio C ÷ d. Using π ≈ 3.14 keeps the arithmetic visible; the calculator's π button is more precise.

“They keep forgetting to double the radius.”
Very common. Encourage a habit: write down d first (d = 2r), then multiply by π. Naming the diameter before calculating prevents most slips.

“Why bother with the graph?”
The straight line through the origin shows that circumference and diameter are in direct proportion, and that its gradient *is* π. It connects circles to the graph work and makes π feel structural, not arbitrary.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.